

Full Length Article

High-resolution image deraining via dual-branch features interaction and fusion

Weilong Huang^a, Jiaxue Mei^a, Tao Yan^{a,*}, Yinghui Wang^a, Xiaojun Chang^{b,c}^a School of Artificial Intelligence and Computer Science, Jiangnan University, Wuxi, 214122, Jiangsu, China^b School of Information Science and Technology, University of Science and Technology, Hefei, 230026, Anhui, China^c Department of Computer Vision, Mohamed bin Zayed University of Artificial Intelligence (MBZUAI), Abu Dhabi, United Arab Emirates

ARTICLE INFO

Keywords:

Interaction network
High resolution image
Frequency-aware
Discrete cosine transform
Image deraining

ABSTRACT

Existing image deraining methods primarily focus on low-resolution images and exhibit inherent limitations when applied to high-resolution images. First, performing feature extraction directly on high-resolution images leads to a significant increase in computational cost and memory consumption. Second, the inherent complexity of high-resolution rainy images, where rain streaks often intermingle with or severely obscure the background, imposes substantial demands on models' capacity to capture both local features and global contextual dependencies. Consequently, many existing models tend to over-smooth high-frequency details during rain removal, resulting in image blurring and the loss of critical structural information. To address these challenges, we propose an efficient Dual-branch Deraining Network (DDNet), which integrates frequency-domain feature interaction and dynamic feature fusion to enhance the representations capacity of its CNN and Transformer branches. Specifically, we propose an Adaptive Frequency Reciprocal Module (AFRM) that adaptively generates frequency spectral filters to enable mutual reinforcement and interaction between the two branches. We further propose a Discrete Cosine Transform (DCT) based Multi-Spectral Fusion Module (MSFM) to dynamically fuse the decoded features from both branches, thereby promoting effective aggregation of multi-spectral information. Moreover, to address the potential feature disparities and uncertain knowledge gaps between the CNN and Transformer architectures, we propose a Dual Differential Cross Attention Module (DDCAM) to mitigate the adverse effects stemming from feature discrepancies. Extensive experiments conducted on both real-world and synthetic high-resolution rainy datasets demonstrate the effectiveness and efficiency of our model. Our code and models are available at: <https://github.com/YT3DVision/HRDerain>.

1. Introduction

Rain streaks serve as a common form of image degradation, often causing visual deterioration issues, such as blurring and occlusion, significantly impacting the performance of subsequent computer vision tasks, including object detection, segmentation and tracking. In modern high-resolution imaging systems, such as 4K (3840 × 2160) high-definition cameras and intelligent surveillance devices, the captured image content becomes richer, and high-frequency details also become more abundant. Nevertheless, rain streaks present more intricate, diverse and refined structures, coupled with a more heterogeneous distribution across the image. Therefore, removing rain streaks from high-resolution images severely challenges existing deep learning-based deraining methods.

In recent years, deep learning-based image deraining methods have achieved significant progress. Deraining methods (Li et al., 2019; Xiao et al., 2022) based on convolutional neural networks (CNNs) and Transformers have emerged and demonstrated excellent performance on several benchmark datasets (Fu et al., 2017; Quan et al., 2021; Yang et al., 2017). However, current mainstream methods are primarily designed for low-resolution images (Zhang & Patel, 2018), and encounter significant challenges when applied to high-resolution images. First, due to the limitations of the Nyquist-Shannon sampling theorem (Nyquist, 1928; Shannon, 1949), low-resolution images inevitably suffer a substantial loss of high-frequency detail information during the downsampling process, whereas the loss of low-frequency information is comparatively minor. Despite the inherent richness of frequency and spatial information in high-resolution images, the increase in image resolution results

* Corresponding author.

E-mail addresses: 6233115012@stu.jiangnan.edu.cn (W. Huang), yantao.ustc@gmail.com (T. Yan).<https://doi.org/10.1016/j.neunet.2026.108936>

Received 8 September 2025; Received in revised form 18 February 2026; Accepted 2 April 2026

Available online 3 April 2026

0893-6080/© 2026 Published by Elsevier Ltd.

in a quadratic growth in feature map size, dramatically increasing both computational complexity and memory consumption. This substantial increase not only significantly complicates model training but also hinders deployment on resource-constrained platforms (Cai et al., 2023a). Secondly, existing methods generally focus on extracting spatial-domain features (Yan et al., 2024), such as local details and global semantic associations, while overlooking the crucial role of frequency-domain information in modeling the image degradation process.

Though several image deraining methods (Jiang et al., 2024; Xiao & Xia, 2025) leverage traditional techniques, such as Fourier Transforms (Cooley et al., 1969), Wavelet Transforms (Mallat, 1989), and Discrete Cosine Transforms (Ahmed et al., 1974) for frequency-domain feature extraction, these methods suffer from inherent limitations. Specifically, their reliance on predefined transform formulas and fixed parameters results in a critical lack of adaptability to diverse image content. Furthermore, the discrepancy between training and testing resolutions results in diminished model generalization (Touvron et al., 2019). While a model pre-trained on low-resolution images is directly applied to high-resolution images, it tends to produce suboptimal recovery results, leading to the amplification of artifact and detail blurring.

In this paper, to exploit the complementary information embedded in the spatial and frequency-domains for high-resolution image deraining, we propose an efficient Dual-branch Deraining Network, named DDNet, which consists of a CNN-based high-resolution branch and a Transformer-based low-resolution branch. Specifically, leveraging the weight-sharing mechanism and local information extraction capabilities of CNNs (LeCun et al., 1998; Luo et al., 2016), the CNN-based high-resolution branch processes high-resolution inputs, focusing on the extraction of local and high-frequency features from the input image. On the other hand, given the high computational complexity of Transformer (Vaswani et al., 2017), the input high-resolution image is first down-sampled and then fed into Transformer-based low-resolution branch for extracting global contextual information. Benefiting from the inherent low-pass filtering property of Transformers (Park & Kim, 2022), the low-resolution branch naturally extracts low-frequency information.

Particularly, to effectively extract and enhance the beneficial frequency information within the dual-branch encoded features, we propose AFRM to adaptively predict spectral filters and selectively refine the frequency components that contribute positively to the restoration of the background. We also propose MSFM to dynamically inject the decoder frequency features of the low-resolution branch into the high-resolution branch for effective frequency feature aggregation. To mitigate the adverse effects of inter-branch feature distribution misalignment, which stems from the inductive biases of CNN (local information modeling) and Transformer (global context aggregation) (Benjdira et al., 2023; Park & Kim, 2022), we propose DDCAM to alleviate these feature disparities. Comprehensive comparisons with other state-of-the-art methods demonstrate the effectiveness and efficiency of our model, as illustrated in Fig. 1.

The main contributions of the work presented in this paper can be summarized as follows:

- First, we propose an efficient Dual-branch Deraining Network (DDNet) for high-resolution image deraining, which integrates frequency-domain feature interaction and dynamic feature fusion to enhance the representations capacity of its CNN and Transformer branches.
- Second, we propose an Adaptive Frequency Reciprocal Module (AFRM) that adaptively generates frequency spectral filters to enable mutual reinforcement and interaction between the two branches.
- Third, we propose a DCT-based Multi-Spectral Fusion Module (MSFM) for information aggregation, and a Dual Differential Cross-Attention Module (DDCAM) for mitigating the adverse effects of feature discrepancies.

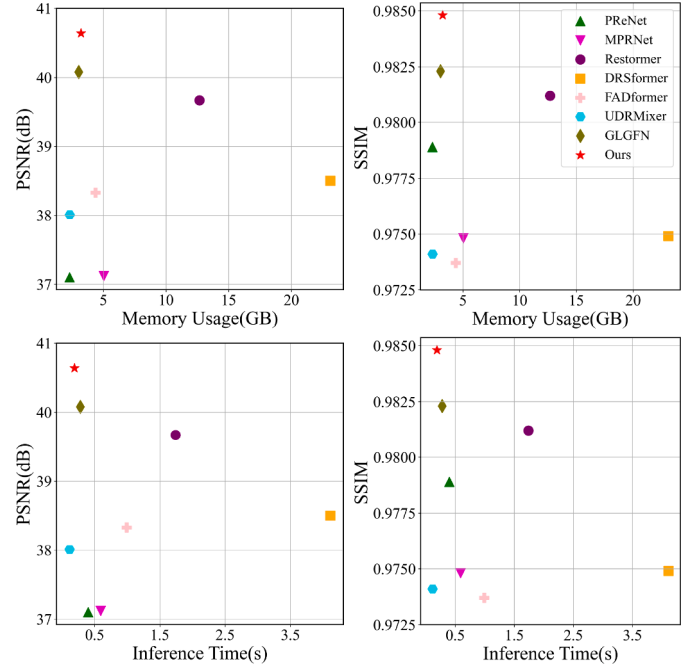


Fig. 1. Our DDNet achieves the best performance on high resolution image deraining task. We show the comparison of our method with existing SOTA methods in terms of computational efficiency (inference time (s) and GPU memory usage (GB)) versus PSNR and SSIM metric on the RealRain-1k-H (Li et al., 2022) dataset. Notably, UDRMixer (Chen et al., 2024a), GLGFN (Yan et al., 2024) and our model are designed specifically for high-resolution image deraining.

- Finally, extensive experiments conducted on real-world and synthetic high-resolution rainy datasets demonstrate that our model outperforms existing state-of-the-art methods with low computational complexity and resource consumption.

2. Related works

In this section, we first review single-image deraining methods, followed by an overview of high-resolution image restoration, and finally a survey of frequency-aware image deraining methods.

2.1. Single-image deraining

single-image deraining aims to remove rain streaks from a single rainy image and restore a rain-free image. Early conventional methods (Gu et al., 2017; Kang et al., 2012; Li et al., 2016; Luo et al., 2015; Zhu et al., 2017) relied on handcrafted prior knowledge, which struggled to accurately model complex rain streak distributions. To overcome the limitations, deep learning methods have emerged, and various CNN-based network architectures, such as multi-scale architectures (Cai et al., 2023b; Jiang et al., 2020; Yasarla & Patel, 2019; Zhou et al., 2023b, 2022), multi-stage architectures (Li et al., 2018; Ren et al., 2019; Zamir et al., 2021), generative adversarial networks (Wang et al., 2025b; Zhao et al., 2024) and diffusion models (Kawar et al., 2022; Luo et al., 2023; Özdenizci & Legenstein, 2023), have been widely employed for image deraining tasks. However, the limited receptive fields (LeCun et al., 1998) of CNNs hinder the ability to extract global information, thus constraining model performance.

With the Transformer demonstrating powerful performance in computer vision (Dosovitskiy et al., 2021), many researchers have been exploring Transformer-based deraining architectures (Chen et al., 2024c; Wang et al., 2025a). For example, Xiao et al. (2022) utilize window-based and spatial-based Transformer architectures to achieve

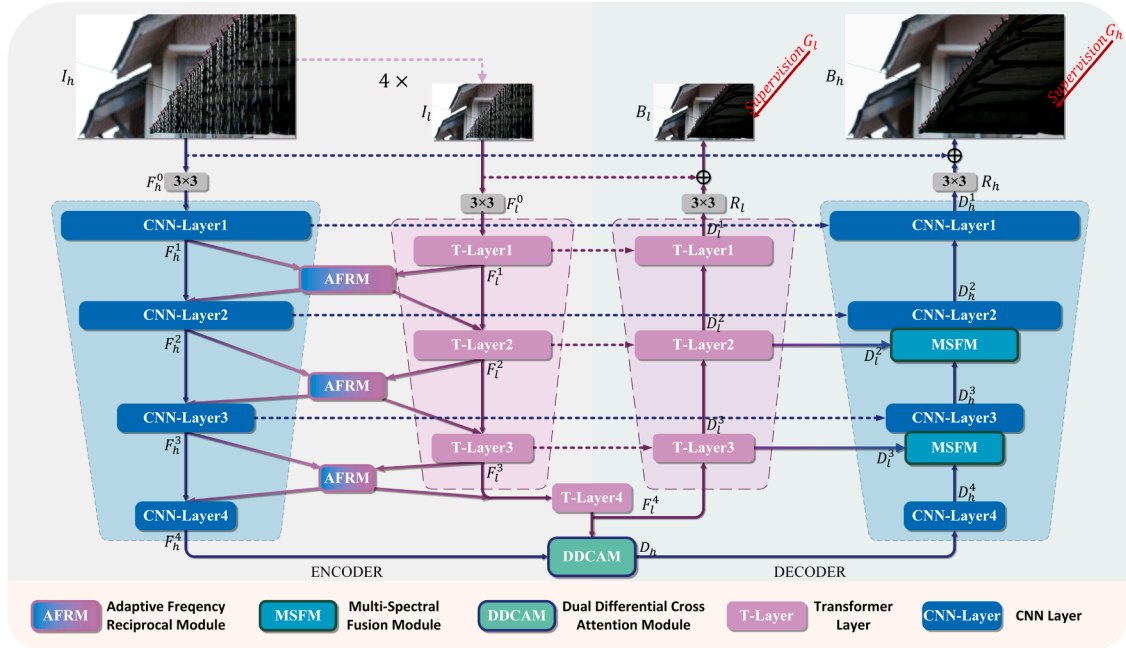


Fig. 2. The architecture of our dual-branch Deraining Network (DDNet).

information complementarity for better background image restoration. However, due to the high computational cost of the attention mechanism (Vaswani et al., 2017), some studies have focused on improving the computational efficiency of Transformers (Tay et al., 2022). Chen et al. (2023a) propose an adaptive sparse selection mechanism to implement a sparse Transformer architecture, filtering redundant features and improving computational efficiency. Zamir et al. (2022) reduce computational costs through a channel-wise transposed attention mechanism.

Furthermore, some studies (Chen et al., 2021; Guo et al., 2022) have focused on developing hybrid models that combine CNN and Transformer architectures. Wang et al. (2024b) propose a multi-scale fusion and decomposition network that combines CNNs and self-attention through an asymmetrical dual-path mutual representation architecture. Similarly, Chen et al. (2023b) propose a stage-wise progressive hybrid framework that combines CNN-based architecture for spatially-varying rain pattern extraction and Transformer-based architecture for global dependency modeling.

Despite the significant improvements achieved by these methods of single-image deraining, their performance often degrades markedly when handling high-resolution images.

2.2. High-resolution image restoration

Compared to existing research on regular resolution image restoration, methods for high-resolution image restoration predominantly focus on image dehazing, with limited exploration in image deraining. Representative works like HR-Dehazer (Bianco et al., 2019) and 4K-Dehazing (Zheng et al., 2021) employ multi-layer convolutional architectures to achieve effective dehazing while maintaining relatively low computational complexity. The EMPF-Net (Wen et al., 2023) employs an encoder-free architecture to achieve efficient remote sensing image dehazing by leveraging multiaxis query learning, supervised cross-scale transposed attention, and a physics-aware intra-layer fusion mechanism. The CINet (Wen et al., 2025a) employs a hierarchical architecture that optimizes skip connections to regulate the information transfer for efficient remote sensing image dehazing. The ADSM (Wen et al., 2025b) utilizes an adaptive self-prompting mechanism to inject degradation and context-aware priors into the diffusion process for efficient all-in-one weather-degraded image restoration. Wang et al. (2024a) con-

struct feature transformations from high-resolution to low-resolution spaces, learning advantageous feature representations from both high-resolution and low-resolution features to achieve high-quality image restoration. PLRR (Xu et al., 2025) proposes a learnable parameterized low-rank regularizer that constrains matrix-factorization factors via neural networks, achieving superior low-rank modeling. Xu et al. (2024) introduce pan-denoising, a regularization paradigm that employs panchromatic weighted representation coefficient total variation to achieve hyperspectral image denoising and enhance downstream classification performance.

To address deraining specifically, Yan et al. (2024) utilize a grafted fusion network, leveraging cross-attention mechanisms and selective kernel fusion to integrate global features into local features, significantly improving high-resolution image deraining performance. However, these methods emphasize spatial domain features while neglecting the analysis and utilization of the frequency information, thereby limiting further improvements in restoration performance.

2.3. Frequency-aware image deraining

Recently, numerous deep learning methods have leveraged frequency information to enhance deraining performance. For example, WaveNet (Zhou et al., 2023a) leverages Discrete Wavelet Transform for efficient feature extraction and employs a Knowledge Distillation mechanism to acquire rich semantic and geometric information from a Transformer teacher network. (Zhang et al., 2024) introduce a wavelet-spatial dual transformer framework, capturing both frequency and spatial features to guide rain streak removal and background recovery. Similarly, based on Wavelet Transforms, Jiang et al. (2024) explore the mutual representation of rain residuals and background components in the frequency domain, aiming to better separate rain layers from background layers and optimize background recovery results. Gao et al. (2024a) design a Fourier-based Transformer framework and incorporate frequency domain contrastive regularization to improve deraining performance. AIRFormer Gao et al. (2024b) introduces a frequency-oriented Transformer that leverages wavelet-based priors and multi-dimensional task-specific queries to achieve an optimal trade-off between restoration quality and inference efficiency across diverse adverse weather conditions.

However, these methods predominantly employ traditional frequency transformation techniques (e.g., Fourier transform (Cooley et al., 1969) and Wavelet Transform (Mallat, 1989)) to statically partition features into fixed frequency bands, lacking adaptive extraction and context-aware modeling of frequency-specific information, which ultimately constrains their restoration capabilities.

3. Our method

3.1. Overall pipeline

In this work, we model rain streaks as an additive degradation. The classic and widely adopted mathematical model for this assumption is:

$$I = B + R, \quad (1)$$

where I is the observed rainy image, B is the clean background image, and R is the rain streak layer. The structure of our Dual-branch Deraining Network (DDNet), as shown in Fig. 2, primarily comprises an asymmetric dual encoder-decoder structure. Specifically, the blue portion represents the high-resolution branch, consisting of multiple multi-scale CNN-based NAF Blocks (Chen et al., 2022) to extract local features and preserve high-frequency information. The pink portion is the low-resolution branch composed of multi-scale Transformer layers with transposed attention (Zamir et al., 2022). The low-resolution branch takes a $4\times$ down-sampled low-resolution image of the original high-resolution rainy image as input to capture global semantic features, and functions as a low-pass filter to retain low-frequency information.

Specifically, given a high resolution rainy image $I_h \in \mathbb{R}^{3 \times H \times W}$ and its downsampled image $I_l \in \mathbb{R}^{3 \times \frac{H}{4} \times \frac{W}{4}}$, where $H \times W$ represents the spatial resolution of the input image. Initially, the two branches separately adopt a 3×3 convolution to extract shallow features $F_h^0 \in \mathbb{R}^{C \times H \times W}$ and $F_l^0 \in \mathbb{R}^{C \times \frac{H}{4} \times \frac{W}{4}}$ from I_h and I_l , where C denotes the feature dimension.

In the high-resolution branch, the shallow feature F_h^0 is fed into a four-level (layer) encoder-decoder structure, where each level consists of N_h^i ($i = 1, 2, 3, 4$) cascaded NAF-Blocks, with the number gradually increasing from shallow to deep layers. Each encoder layer performs spatial dimension reduction and channel expansion to extract deep features while reducing computational complexity.

Similarly, in the low-resolution branch, the shallow feature F_l^0 is fed into a three-level encoder-decoder architecture, with each level consisting of N_l^j ($j = 1, 2, 3$) cascaded Transformer blocks. Since the input of the low-resolution branch is derived from the downsampled rainy image, spatial dimension reduction and channel expansions are only executed at the second and third levels to mitigate information loss caused by excessive feature downsampling.

The adaptive Frequency Reciprocal Module (AFRM), detailed in Section 3.2, takes the encoded features F_h^i, F_l^i ($i = 1, 2, 3$) from the same level of both encoder branches as inputs. By generating adaptive spectral filters, it can enhance the frequency information conveyed within dual-branch features.

At the bottleneck of our DDNet, the encoded feature F_l^3 from the low-resolution branch is fed into T-Layer4, which comprises several Transformer blocks and produces the enhanced feature F_l^4 . Then, a Dual Differential Cross Attention Module (DDCAM), which takes F_l^4 and the encoded feature F_h^4 as input, integrates dual-branch features to mitigate inter-branch discrepancy and imbalance. A comprehensive elaboration on DDCAM is provided in Section 3.3. Then, at the decoder stage, our DDNet performs spatial dimension expansion and channel reduction from deeper layers to shallower layers for progressively reconstructing multi-scale rain streak features in both high-resolution and low-resolution. Notably, in the high-resolution branch of the decoder, two DCT-based Multi-Spectral Fusion Modules (MSFMs), detailed in Section 3.4, take the decoded feature pairs (D_h^4, D_l^3) and (D_h^3, D_l^2) as input to generate fused features at the 3rd and 2nd scales, respectively. These fused features are then fed into the corresponding CNN-based decoder layers for further feature reconstruction.

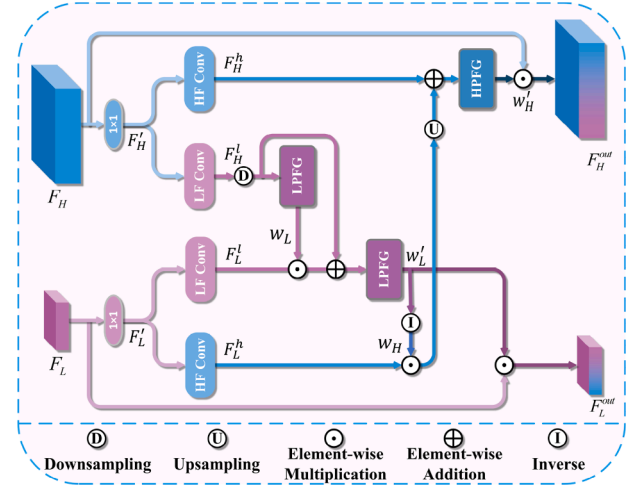


Fig. 3. The structure of adaptive frequency reciprocal module (AFRM).

To ensure training stability, we also adopt skip connections between encoders and decoders at corresponding scales for both branches. Finally, the decoded features D_h^i and D_l^i from both branches are processed by a 3×3 convolution to aggregate information across channels, generating rain residual feature maps R_h, R_l , both of which are then added to I_h and I_l to obtain the derained results B_h, B_l , respectively.

3.2. Adaptive frequency reciprocal module (AFRM)

As analyzed in Section 1, the high-resolution branch preserves a comprehensive spectral representation, whereas the low-resolution branch manifests a severe deficiency of high-frequency components. To further leverage the differential information between the two branches, we propose an Adaptive Frequency Reciprocal Module (AFRM), as illustrated in Fig. 3. Our AFRM can effectively extract and enhance frequency information that contributes to the restoration of the rain-free image from the dual-branch encoded features. Moreover, it can also mitigate the adverse effects caused by the occlusion and blurring phenomena of rain streaks on the extraction and enhancement of frequency components.

Our AFRM comprises two types of core components: frequency-aware convolutions, including LF Conv (Low-Frequency Convolution) and HF Conv (High-Frequency Convolution), and filter generators, including LPFG (Low Pass Filter Generator) and HPFG (High Pass Filter Generator). The LF Conv employs 5×5 convolutional layers to adaptively extract preliminary low-frequency features, utilizing their large receptive fields to encompass broader contextual information and offering excellent smoothing effects. In contrast, the HF Conv employs 3×3 convolutional layers to adaptively extract preliminary high-frequency features, utilizing their small receptive fields to capture intricate local details and providing strong edge-enhancement effects. While we acknowledge the absence of a strictly analytical derivation for this specific configuration, the selection was determined via a semi-empirical approach, integrating extensive experimental validation with foundational theoretical insights. The LPFG is implemented by a kernel-wise softmax operation (Chen et al., 2024b; Zou et al., 2023), which can effectively suppress high-frequency noise interference by smoothing high-level features. Conversely, the HPFG is implemented by inverting the operation, which is able to enhance the high-frequency background details. AFRM employs filter generators (Chen et al., 2024b) to generate adaptive spectral filters from preliminary features, thereby enhancing robustness across diverse scenarios. Therefore, AFRM is able to adaptively enhance beneficial high-frequency information to preserve fine-grained background details and strengthen low-frequency information to maintain overall structural stability simultaneously.

Specifically, given the feature $\mathbf{F}_H \in \mathbb{R}^{C_1 \times \tilde{H} \times \tilde{W}}$ from the high-resolution branch and the feature $\mathbf{F}_L \in \mathbb{R}^{C_2 \times \frac{\tilde{H}}{2} \times \frac{\tilde{W}}{2}}$ from the low-resolution branch, AFRM first aligns the channel dimensions of both feature maps to $\tilde{C}$ via a 1×1 convolution, obtaining $\mathbf{F}'_H$ and $\mathbf{F}'_L$. Subsequently, the processing within this module can be divided into high-frequency information processing pathway (marked in blue color) and low-frequency information processing pathway (marked in magenta color).

The low-frequency information processing pathway within the AFRM is structured as a two-stage enhancement mechanism. Since the input to the low-resolution branch is directly downsampled from the original image, its spectral diversity is constrained, leading to attenuated high-frequency details and degraded low-frequency components. To compensate these spectral deficiencies, AFRM employs LF Convs (5×5) to extract initial low-frequency features, $\mathbf{F}_1^H$ and $\mathbf{F}_1^L$ from $\mathbf{F}'_H$ and $\mathbf{F}'_L$, respectively. Our first stage addresses the deficiency in $\mathbf{F}_1^L$, which means the $\mathbf{F}_1^H$ feature is first downsampled and fed into a LPFG (Low Pass Filter Generator) to produce the first low-frequency filter (w_L). This filter is then applied to $\mathbf{F}_1^L$ via element-wise multiplication, providing a crucial informational complement (enhancement) using guidance from the high-resolution features. Subsequently, the filtered feature is summed with $\mathbf{F}_1^H$ to obtain the fused feature. Then, in the second enhancement stage, this fused feature is fed into another LPFG to produce the final filter (w'_L), which is ultimately applied to the input feature $\mathbf{F}_L$ to produce the final low-frequency output feature map $\mathbf{F}_{out}^L$. This design ensures that the low-resolution branch features undergo two successive filtering operations (one for cross-branch guidance, one for final enhancement), whereas the high-resolution branch features undergo filtering only in the final high-frequency pathway stage.

The high-frequency information processing pathway adheres to a similar methodology. Initially, an inverse operation is applied to w'_L to obtain the first high-frequency filter w_H . Subsequently, the high-frequency information extracted from the low-resolution features, termed $\mathbf{F}_1^H$, is filtered by w_H to preliminarily enhance the high-frequency information. Then, the filtered feature is upsampled and summed with the high-frequency features extracted from the high-resolution feature, termed $\mathbf{F}_H^h$, to produce the fused high-frequency feature map. Finally, a HPFG is applied to the fused feature to produce the second high-frequency filter w'_H , which is then applied to the original input feature $\mathbf{F}_H$ to obtain the high-frequency output feature $\mathbf{F}_H^{out}$.

We employ a kernel-wise softmax operation to implement the low-pass filter. Specifically, given a feature $\tilde{\mathbf{F}} \in \mathbb{R}^{K^2 \times \tilde{H} \times \tilde{W}}$, where K (default set to 5) represents the kernel size and $\tilde{H} \times \tilde{W}$ represents the spatial resolution, the kernel-wise softmax can be defined as:

$$\text{Soft}_L(\tilde{\mathbf{F}}) = \text{Softmax}(\tilde{\mathbf{F}}_{i,j}) = \frac{\exp(\tilde{\mathbf{F}}_{i,j}^{p,q})}{\sum_{p,q \in \Phi} \exp(\tilde{\mathbf{F}}_{i,j}^{p,q})}, \quad (2)$$

where $\tilde{\mathbf{F}}_{i,j}^{p,q}$ represents the weight at position (p, q) in the filter kernel corresponding to feature coordinates (i, j) of $\tilde{\mathbf{F}}$, and Φ represents the scale of the filter kernel, i.e., $K \times K$. Similar to Huang et al. (2023), Magid et al. (2021), in order to obtain high-pass filter, we first generate a low-pass filter using a kernel-wise softmax with its kernel size set to 3, then we subtract it from an identity kernel E , the process can be defined as follows:

$$\text{Soft}_H(\tilde{\mathbf{F}}) = E - \text{Soft}_L(\tilde{\mathbf{F}}) = E - \frac{\exp(\tilde{\mathbf{F}}_{i,j}^{p,q})}{\sum_{p,q \in \Phi} \exp(\tilde{\mathbf{F}}_{i,j}^{p,q})}. \quad (3)$$

3.3. Dual differential cross attention module (DDCAM)

The inherent representational discrepancies between CNN-based and Transformer-based feature extractors (Benjdira et al., 2023; Park & Kim, 2022) may negatively impact the model's performance. Thus, the Dual Differential Cross Attention Module (DDCAM), as shown in Fig. 4, is proposed to mitigate the feature distribution discrepancies between dual-

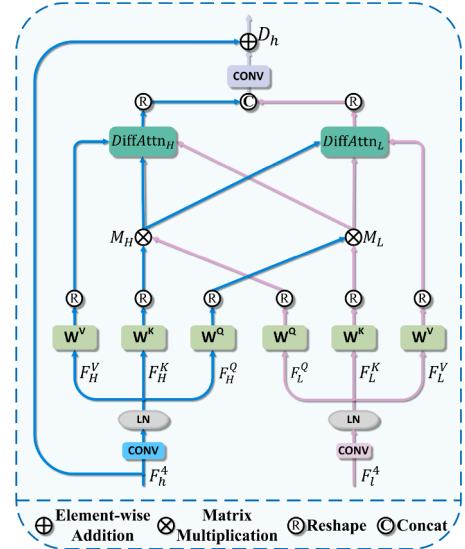


Fig. 4. The structure of Dual Differential Cross Attention Module (DDCAM).

branches before the fusion stage. It takes dual-branch features, $\mathbf{F}_H^4$ and $\mathbf{F}_L^4$, as inputs.

First, the corresponding queries, keys, and values are obtained from the inputs, taking $\mathbf{F}_H^Q$ as an example, which is defined as follows:

$$\mathbf{F}_H^Q = \text{reshape}(W_H^Q(\text{LN}(\text{Conv}_{1 \times 1}(\mathbf{F}_H^4))), \quad (4)$$

where $\text{LN}(\cdot)$ denotes layer normalization (Ba et al., 2016). $\text{Conv}_{1 \times 1}(\cdot)$ represents a 1×1 convolution. W_H^Q represents the mapping matrix, and $\text{reshape}(\cdot)$ denotes the feature reshape operation. The $\mathbf{F}_H^K$, $\mathbf{F}_H^V$ and $\mathbf{F}_L^K$, $\mathbf{F}_L^V$ can be obtained similarly. Subsequently, the module calculates the attention maps M_H and M_L via two symmetric cross-attention mechanisms (Lu et al., 2019), which can enhance the similarity between two features and reduce the disparity in feature distributions. Furthermore, for computational efficiency, we employ transposed attention (Zamir et al., 2022) to reduce computational complexity.

Notably, since the computation of the two attention maps is sensitive to noisy and redundant tokens, which arise from feature discrepancies between CNN and Transformer, a differential strategy is introduced to decrease attention noise and prevent over-attention to irrelevant feature representations. Specifically, inspired by noise-canceling headphones and differential amplifiers in electronic engineering (Laplante et al., 2018), Ye et al. (2025) proposed a differential attention mechanism to calculate the difference between two attention maps for eliminating noise in attention computation. Based on this research, two differential attention maps, named DiffAttn_H and DiffAttn_L , are obtained by leveraging the differential strategy formula as follows:

$$\begin{aligned} \text{DiffAttn}_H &= [\text{Softmax}(M_H) - \lambda_H \text{Softmax}(M_L)] \mathbf{F}_H^V, \\ \text{DiffAttn}_L &= [\text{Softmax}(M_L) - \lambda_L \text{Softmax}(M_H)] \mathbf{F}_L^V, \\ \text{s.t. } \lambda_H &= \exp(\lambda_{q_1} \cdot \lambda_{k_1}) - \exp(\lambda_{q_2} \cdot \lambda_{k_2}) + \lambda_{init}, \\ \lambda_L &= \exp(\lambda_{q_2} \cdot \lambda_{k_2}) - \exp(\lambda_{q_1} \cdot \lambda_{k_1}) + \lambda_{init}, \end{aligned} \quad (5)$$

where $\lambda_{q_1}, \lambda_{k_1}, \lambda_{q_2}, \lambda_{k_2}$ are learnable vectors. λ_{init} is a constant weighting parameter, defined as $\lambda_{init} = 0.8 - 0.6 \times \exp(-0.3 \cdot (l - 1))$, where $l = 5$ indicates the layer index. The formulas for computing λ_H and λ_L are adopted from Ye et al. (2025).

Finally, the output of the DDCAM can be obtained by reshaping, channel concatenation and a residual connection, as follows:

$$\mathbf{D}_h = \text{Conv}(\text{Concat}(\text{reshape}(\text{DiffAttn}_H) + \text{reshape}(\text{DiffAttn}_L))) + \mathbf{F}_H. \quad (6)$$

3.4. DCT-based multi-spectral fusion module (MSFM)

Global Average Pooling (GAP) offers the advantage of computational efficiency, facilitating the rapid capture of global information for Channel Attention mechanisms (Hu et al., 2018). However, this efficiency inherently restricts its capacity to extract complex information for various inputs. To address this issue, Qin et al. (2021) demonstrate that “GAP is a special case of the Discrete Cosine Transform, with its value proportional to the lowest frequency component of the Discrete Cosine Transform (Ahmed et al., 1974)” (detailed proof is provided in our supplementary material), and propose the multi-spectral channel attention mechanism et al. (Qin et al., 2021). Building upon previous research, we propose MSFM, which is illustrated in the Fig. 5. MSFM employs pre-selected distinct Discrete Cosine Transform (DCT) spectral components to extract frequency information from decoded features of dual-branch, and dynamically fuse them thereby facilitating frequency domain information aggregation. By employing MSFM, our method can fully exploit the frequency information within features, which contributes to a more accurate and efficient fusion process.

Specifically, given input features $\hat{\mathbf{F}}_H \in \mathbb{R}^{\hat{C}_1 \times \hat{H} \times \hat{W}}$ and $\hat{\mathbf{F}}_L \in \mathbb{R}^{\hat{C}_2 \times \hat{H} \times \hat{W}}$, the two features are concatenated along the channel dimension to generate feature $\hat{\mathbf{F}} \in \mathbb{R}^{(\hat{C}_1 + \hat{C}_2) \times \hat{H} \times \hat{W}}$. Subsequently, $\hat{\mathbf{F}}$ is equally divided into n parts along the channel dimension, termed $\hat{\mathbf{F}}^k = \text{Split}(\hat{\mathbf{F}})$, and $k = 0, 1, \dots, n-1$. For each part, frequency information is extracted using DCT-based channel information aggregation operation, thus generating a channel weight vector $\mathbf{V}_{\text{freq}}$. Subsequently, a fully connected layer is used to compress and expand the channel dimension of $\mathbf{V}_{\text{freq}}$, facilitating information interaction between different channels. To generate the fused feature, element-wise multiplication is performed between $\hat{\mathbf{F}}$ and the expanded channel weight vector along the channel dimension, followed by a 1×1 convolution to compress the channel dimension to $\hat{C}_1$. Finally, $\hat{\mathbf{F}}_H$ is then added back to the fused feature as a residual, resulting in the output feature $\hat{\mathbf{F}}_{\text{fuse}}$. The overall process of the DCT fusion module can be defined as follows:

$$\begin{aligned} \mathbf{V}_{\text{freq}} &= \text{DCT}_{\text{Trans}}(\hat{\mathbf{F}}), \\ \hat{\mathbf{F}}_{\text{fuse}} &= \text{Conv}_{1 \times 1}(\text{FC}(\mathbf{V}_{\text{freq}}) \cdot \hat{\mathbf{F}}) + \hat{\mathbf{F}}_H, \end{aligned} \quad (7)$$

where $\text{FC}(\cdot)$ represents the function of the fully connected layer. $\text{DCT}_{\text{Trans}}(\cdot)$ refers to our proposed DCT-based channel information aggregation operation, which is defined as:

$$\begin{aligned} D_{i,j}^{u,v} &= \cos\left(\frac{\pi u}{H}\left(i + \frac{1}{2}\right)\right) \cos\left(\frac{\pi v}{W}\left(j + \frac{1}{2}\right)\right), \\ \mathbf{V}^k &= \sum_{i=0}^{H-1} \sum_{j=0}^{W-1} \hat{\mathbf{F}}_{i,j}^k D_{i,j}^{u,v}, \end{aligned} \quad (8)$$

$$\text{DCT}_{\text{Trans}}(\hat{\mathbf{F}}) = \text{Concat}(\mathbf{V}^k),$$

s.t. $k \in \{0, 1, \dots, n-1\}$, $(u, v) \in \Omega$,

where $D_{i,j}^{u,v}$ represents the 2-D DCT basis function. $\text{Concat}(\cdot)$ denotes channel concatenation operation. Ω represents the preselected sequence of frequency component coordinates, and the selection criteria are detailed subsequently.

Our selection criterion is inspired by JPEG image compression theory (Wallace, 1992). In JPEG encoding, Discrete Cosine Transform (DCT) is used to analyze the high and low-frequency information content of an image for retaining salient components while discarding perceptually insignificant components. The resulting DCT spectrum exhibits a structured distribution: low-frequency information concentrates in the upper-left corner, while high-frequency details dominate the lower-right region. A quantization table then suppresses perceptually insignificant high-frequency coefficients by setting them to zero to achieve compression. Finally, the ZIGZAG scan traverses the DCT block in order of increasing spatial frequency, as illustrated by the orange path in the numerical matrix of Fig. 5, prioritizing salient components for entropy coding. In our method, given that the feature $\hat{\mathbf{F}}$ is formed by concatenating features from high-resolution and low-resolution branches along

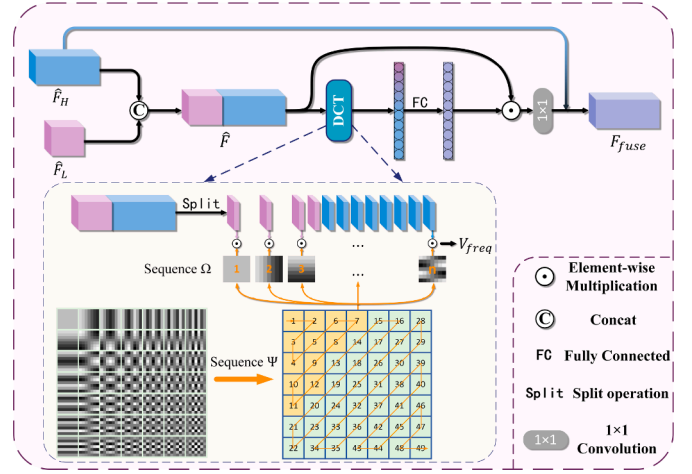


Fig. 5. The structure of DCT-based multi-spectral fusion module (MSFM).

the channel dimension, its leading features exhibit rich low-frequency information, while the trailing features are characterized by rich high-frequency information, which properly aligns with the ZIGZAG scan algorithm. Therefore, the ZIGZAG scan algorithm is employed to number different DCT frequency components. Specifically, we sequentially arrange these frequency components, forming a sequence Ψ , and select the top- n elements to constitute sequence Ω . The specific details are illustrated in the highlighted DCT spectrum and numerical matrix within the light-colored bounding box of Fig. 5. It should be noted that when $n = 1$, the $\text{DCT}_{\text{Trans}}(\cdot)$ operation degenerates into the GAP operation.

3.5. Loss function

First, L_1 loss is employed to supervise the outputs $\mathbf{B}_h$ and $\mathbf{B}_l$ of the network at both resolutions, formulated as follows:

$$\begin{aligned} L_h &= \|\mathbf{B}_h - \mathbf{G}_h\|_1, \\ L_l &= \|\mathbf{B}_l - \mathbf{G}_l\|_1, \end{aligned} \quad (9)$$

where $\mathbf{G}_h$ and $\mathbf{G}_l$ represent the rain-free ground truth image and its downsampled counterpart, respectively.

Subsequently, given that the primary output of our network is the high-resolution output, SSIM loss is adopted solely to supervise $\mathbf{B}_h$, defined as follows:

$$L_s = 1 - \text{SSIM}(\mathbf{B}_h, \mathbf{G}_h). \quad (10)$$

In our work, L_1 loss is used to reduce pixel-level intensity errors for cleaner results, while SSIM loss preserves structural details and texture consistency to enhance visual quality. The final loss of our network is obtained by a weighted sum of L_1 and L_{ssim} , described as:

$$L_{\text{total}} = \alpha \cdot L_h + \beta \cdot L_l + \gamma \cdot L_s, \quad (11)$$

where α , β and γ are hyper-parameters.

4. Experiments

4.1. Datasets for Experiments

We conduct experiments on two real-world datasets, including the real-world high-resolution rainy dataset LHP-Rain (Guo et al., 2023) and RealRain1K (Li et al., 2022), which comprises two subsets: RealRain1K-H (heavy rain version) and RealRain1K-L (light rain version). In addition, we also employ another synthetic high-resolution rainy dataset RainDirection (Liu et al., 2021). A detailed description is provided in the Table 1.

Table 1

Detail presentation of high-resolution rainy datasets. “Real” and “Syn” denote the real-world and synthetic datasets.

Dataset	Train/Val/Test	Avg. Res	Category
RealRain-1k-L	784/112/224	1512 * 973	Real
RealRain-1k-H	784/112/224		
LHP-Rain	2,100/600/300	1920 * 1080	Real
RainDirection	2,920/ - /430	1946 * 1444	Syn

4.2. Implementation details

DDNet employs a dual-branch architecture. In the high-resolution branch, a four-level architecture is utilized, where the number of base modules N_h^i ($i = 1, 2, 3, 4$) in each level is sequentially set to $[1, 1, 4, 8]$. Conversely, the low-resolution branch, which comprises a three-level architecture, has the number of base modules N_l^j ($j = 1, 2, 3$) set to $[1, 2, 2]$. The number of Transformer blocks N_j^4 in the low-resolution bottleneck layer is set to 2. The initial channel number C for both branches is uniformly set to 32. Furthermore, we set the hyper-parameter n to 16 within Multi-Spectral Fusion Module, thus selecting the top 16 frequency components to construct the sequence Ω . The hyper-parameters α, β and γ in Eq. 11 are set to 1, 0.5, and 0.2, respectively. The values of α, β and γ were determined empirically through experimental analysis. These coefficients are intended to appropriately scale each component and maintain a balanced overall contribution. As the high-resolution branch exhibits greater importance in the reconstruction process, its corresponding weight is assigned a relatively larger value. The entire training and testing procedure for the network is conducted on the PyTorch platform, utilizing an NVIDIA GeForce RTX3090 GPU with 24GB of memory. The AdamW optimizer ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay $1e^{-4}$) and a combination of L_1 loss and SSIM loss are employed to train our model. The initial learning rate is set to $3e^{-4}$ and gradually reduced to $1e^{-6}$ with the cosine annealing (Loshchilov & Hutter, 2017). To mitigate the negative impact of excessive resolution discrepancies between training and testing on model performance, we adopt a progressive learning strategy (Zamir et al., 2022). Specifically, our model begins training with smaller patch sizes and gradually increases the patch size as the training progresses. The total number of training epochs is set to 1000, with the training patch size and batch size pairs configured as follows: $[(128^2, 24), (256^2, 12), (384^2, 5), (512^2, 3), (640^2, 2), (768^2, 1)]$. The number of training epochs for each patch size is $[200, 200, 150, 200, 150, 100]$, respectively.

4.3. Comparison with state-of-the-arts methods

In this work, we select 9 representative image deraining methods, which consist two CNN-based methods, PReNet, MPRNet (Ren et al., 2019; Zamir et al., 2021); three Transformer-based methods, IDT, Restormer, DRSformer, (Chen et al., 2023a; Xiao et al., 2022; Zamir et al., 2022); two hybrid-based methods, HCT-FFN, GLGFN (Chen et al., 2023b; Yan et al., 2024) and four frequency-aware methods, FADFormer, UDRMixer (Chen et al., 2024a; Gao et al., 2024a), UHD-Processor (Liu et al., 2025) and ERR (Zhao et al., 2025) for comparison with our model. Among these methods, GLGFN (Yan et al., 2024), UDRMixer (Chen et al., 2024a), UHD-Processor (Liu et al., 2025) and ERR (Zhao et al., 2025) are designed specifically for high-resolution image deraining. To maintain the brevity of the manuscript, detailed hyperparameter settings of each method were omitted. However, we confirm that all comparative experiments were strictly conducted using the exact hyperparameters reported in their official implementations.

4.3.1. Quantitative evaluation

Table 2 shows the quantitative metrics, parameters and FLOPs of our proposed methods and various comparison methods on two real-world high resolution datasets RealRain1k (Li et al., 2022), LHP-Rain (Guo

et al., 2023) and a synthetic dataset RainDirection (Liu et al., 2021). Peak Signal-to-Noise Ratio (PSNR) and Structural Similarity Index Measure (SSIM) are calculated using three color channels in RGB space.

It is evident that our method DDNet achieves the best quantitative metrics across all datasets compared to other competitive methods. Taking the PSNR as an example, in the heavy rain version of the real-world dataset RealRain1k (Li et al., 2022), DDNet outperforms the sub-optimal method GLGFN (Yan et al., 2024) by 0.56 dB. This advantage is even more pronounced in the light rain version, where our method surpasses the sub-optimal method by 0.83 dB. On the LHP-Rain dataset (Guo et al., 2023), which has a higher average resolution compared to RealRain1k (Li et al., 2022), our DDNet slightly outperforms GLGFN (Yan et al., 2024), but shows a significant improvement over Restormer (Zamir et al., 2022) and DRSformer (Chen et al., 2023a), with 0.36 dB and 0.3 dB higher, respectively. In the synthetic dataset RainDirection (Liu et al., 2021), our method achieves 0.1 dB higher PSNR than Restormer. As detailed in Table 2, our method also achieves the highest scores in SSIM. To maintain conciseness, a detailed analysis of this metric is intentionally excluded here. On the other hand, in terms of parameters and computational complexity, tested with image patch sizes of 256×256 , our method achieves a lightweight architecture with 12.5M parameters and 18.55G FLOPs when operating on 256×256 image patches, achieving state-of-the-art performance while maintaining lightweight characteristics.

Several methods exhibit suboptimal performance, CNN-based methods PReNet (Ren et al., 2019) and MPRNet (Zamir et al., 2021) are limited by the receptive field of CNNs, while IDT (Xiao et al., 2022) and HCT-FFN (Chen et al., 2023b) simply concatenate global and local information in a sequential manner, which impacts the performance severely. In addition, FADFormer (Gao et al., 2024a), UDRMixer (Chen et al., 2024a), UHD-Processor (Liu et al., 2025) and ERR (Zhao et al., 2025) employ the Fourier Transform or DCT as frequency analysis tool, which lacks adaptive capabilities and consequently impedes their performance.

4.3.2. Qualitative evaluation

For the sake of clarity and to facilitate reading, we limit the presentation of visualization results to selected methods that yielded the optimal performance. Additional experimental results are provided within the supplementary material. To evaluate the qualitative performance, Fig. 6 presents the deraining results of DDNet and other state-of-the-art comparison methods on the real-world dataset RealRain1k (Li et al., 2022). The first scene illustrates the deraining results under the heavy rain version of this dataset. As highlighted in the zoomed-in regions, our method successfully removes all rain streaks while preserving the most comprehensive overall structural information and texture details. In contrast, UHD-Processor (Liu et al., 2025), ERR (Zhao et al., 2025), UDRMixer (Chen et al., 2024a), GLGFN (Yan et al., 2024), and Restormer (Zamir et al., 2022) fail to eliminate rain streaks, while DRSformer (Chen et al., 2023a) and MPRNet (Zamir et al., 2021) significantly degrade the background information. The second scene in Fig. 6 shows the deraining results under the light rain version. Substantial rain streaks indicated by the red rectangles are retained by several methods, whereas DDNet completely removes rain streaks and successfully recovers the background information. The green rectangles showcase the deraining performance on complex textured backgrounds, where DDNet demonstrably achieves the best visual quality.

Fig. 7 displays the deraining results on the real-world dataset LHP-Rain (Guo et al., 2023). Due to the increased complexity of the image backgrounds and the dense rain streaks, most methods fail to eliminate rain streaks in the first scene. Additionally, the second scene specifically highlights the degradation of the image background caused by heavy rain-induced splashes and rain streaks on surfaces. As observed in the green rectangles, several methods fail to recover the obscured background information. In comparison, our method removes the vast

Table 2

Quantitative results of different method on real-world high-resolution datasets RealRain-1K (Li et al., 2022), LHP-rain (Guo et al., 2023) and synthetic dataset RainDirection (Liu et al., 2021). Best results are bolded. FLOPs is calculated on 256×256 patch.

Methods	Publication	RealRain-1k-H		RealRain-1k-L		LHP-Rain		RainDirection		Param(M)↓	FLOPs(G)↓
		PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑	PSNR↑	SSIM↑		
PreNet	CVPR19	37.10	0.9789	38.51	0.9822	33.26	0.9333	28.34	0.8878	0.17	66.25
MPRNet	CVPR21	37.12	0.9748	39.86	0.9833	33.64	0.9386	29.67	0.8977	3.64	548.65
IDT	TPAMI22	32.80	0.9632	34.45	0.9718	33.34	0.9338	28.48	0.8925	16.39	58.44
Restormer	CVPR22	39.67	0.9812	40.91	0.9851	33.77	0.9373	30.69	0.9121	26.10	140.99
DRSformer	CVPR23	38.50	0.9749	39.98	0.9812	33.83	0.9400	30.57	0.9050	33.63	220.38
HCT-FFN	AAAI23	32.64	0.9587	34.30	0.9686	33.24	0.9318	29.15	0.9072	0.87	80.25
FADFormer	ECCV24	38.33	0.9737	39.41	0.9800	33.44	0.9346	30.42	0.9064	6.96	48.51
UDRMixer	ARXIV24	38.01	0.9741	39.03	0.9787	33.50	0.9369	30.14	0.9036	4.86	15.71
GLGFN	TCSVT24	40.08	0.9823	41.04	0.9854	34.10	0.9448	30.62	0.9090	31.64	31.62
UHD-Processor	CVPR25	36.49	0.9750	37.47	0.9803	33.47	0.9401	29.96	0.8904	1.60	4.17
ERR	CVPR25	38.35	0.9813	39.46	0.9844	33.49	0.9398	30.12	0.8987	1.13	2.41
DDNet(ours)	-	40.64	0.9848	41.87	0.9875	34.13	0.9450	30.79	0.9128	12.50	18.55

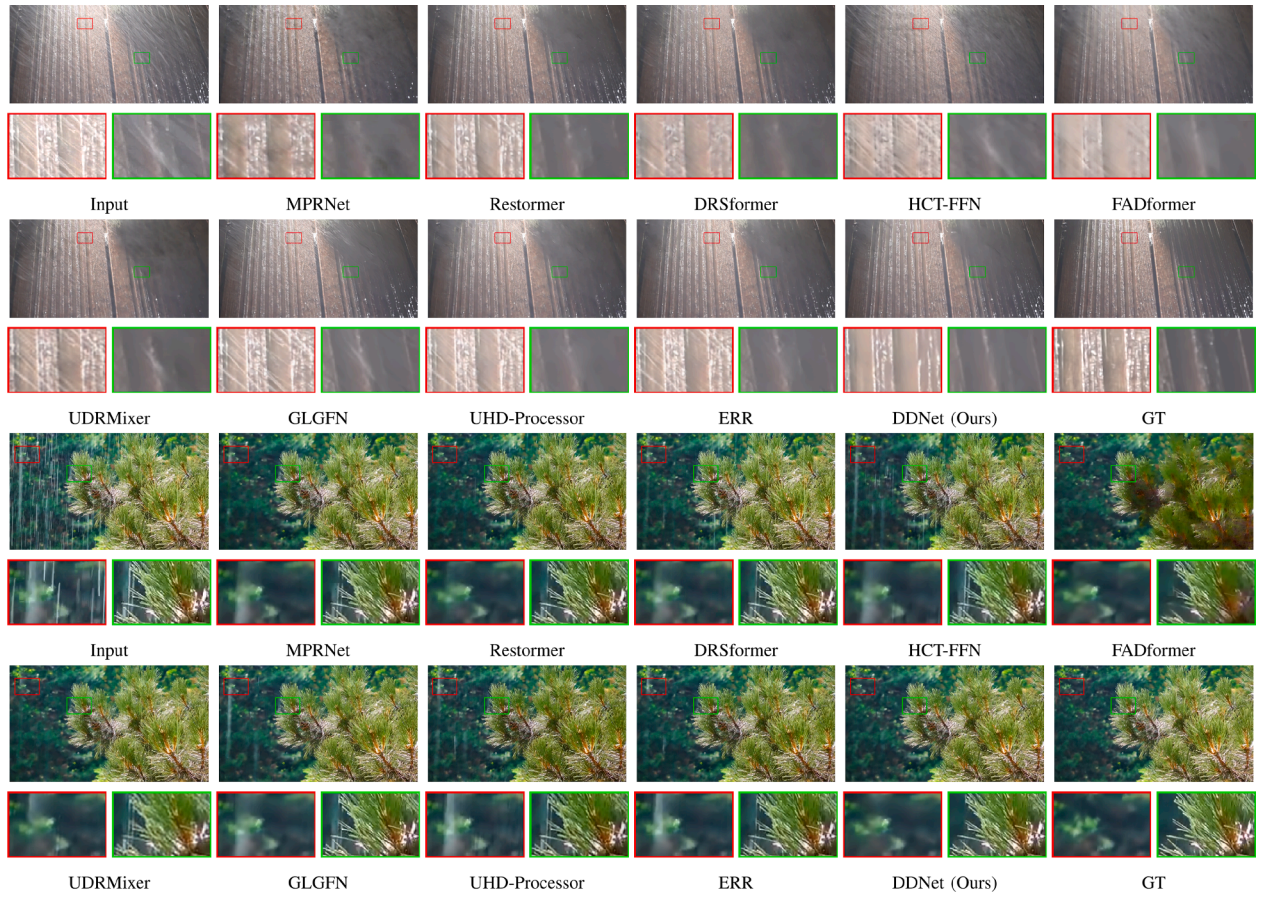


Fig. 6. The visual results obtained by different methods from real-world high-resolution dataset RealRain1k (Li et al., 2022). The first scene is from RealRain1k-H (heavy rain version), and the second scene is from RealRain1k-L (light rain version).

majority of rain streaks and successfully recovers the background information occluded by rain streaks.

Finally, on the synthetic dataset RainDirection (Liu et al., 2021), where the rain streak patterns are uniform, the recovery results of some methods are relatively similar. Therefore, we only present one scene in Fig. 8. Additional visualization results are exhibited in the supplementary material. The PSNR and SSIM values of each result are also marked. As shown in the red and green rectangles, UHD-Processor (Liu et al., 2025), ERR (Zhao et al., 2025), Restormer (Zamir et al., 2022), GLGFN (Yan et al., 2024) and other methods exhibit obvious residual rain streaks. HCT-FFN (Chen et al., 2023b) shows a clear color deviation from the ground truth, as highlighted by the green rectangular an-

notations. In contrast, our network removes the majority of rain streaks while preserving original color information, resulting in superior visual fidelity.

4.4. Ablation study

In this subsection, we evaluate various variants of our proposed DDNet on real-world datasets RealRain1K-H (Li et al., 2022) to validate the effectiveness of our dual-branch architecture and three key modules, including AFRM, MSFM, and DDCAM. Furthermore, we also conduct an ablation study on the parameter n within the MSFM.



Fig. 7. The visual results obtained by our method and other methods on real-world high-resolution dataset LHP-Rain (Guo et al., 2023).

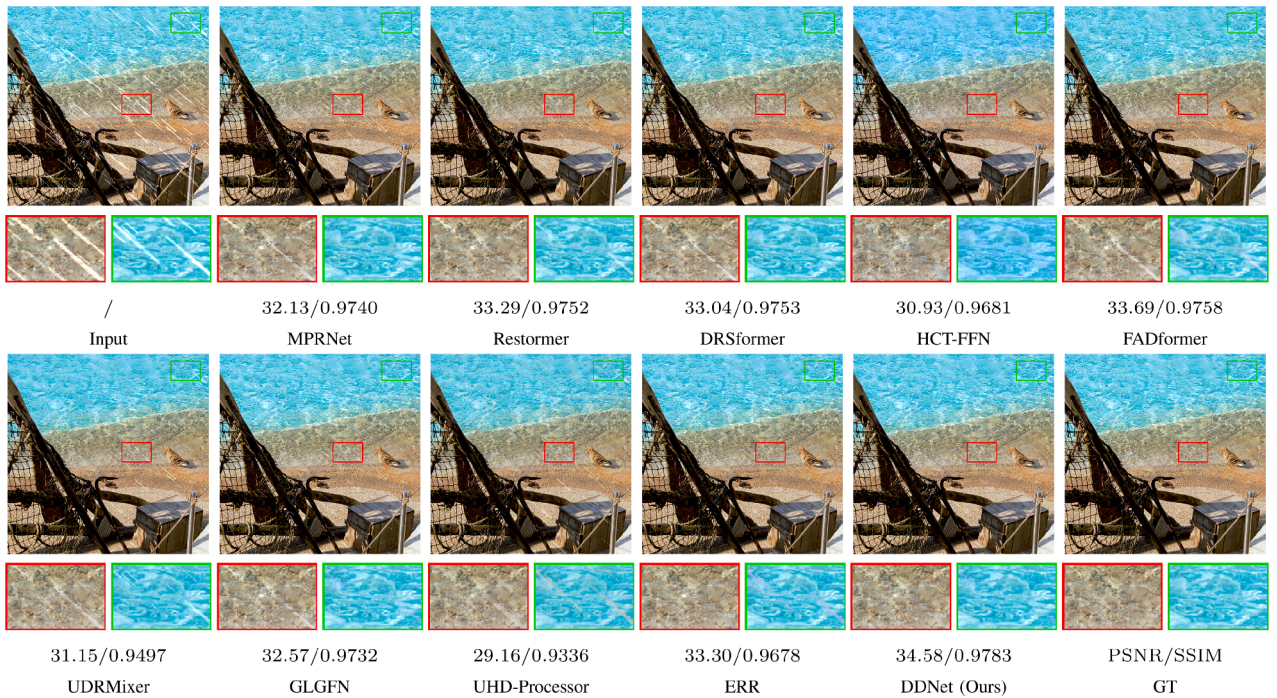


Fig. 8. The visual results and PSNR/SSIM metrics obtained by different methods from synthetic high-resolution dataset RainDirection (Liu et al., 2021).

Table 3

Ablation study on three key modules of DDNet. “DDCAM $\rightarrow$ CA” denotes the substitution of the DDCAM module with a standard cross-attention (CA) mechanism. The $\checkmark$ of “Fusion Strategy” column indicates which fusion strategy is employed: sum, concat (cat), or MSFM. The $\checkmark$ and $\times$ of the “w/o AFRM” column denote “with AFRM” and “without AFRM”, respectively. FLOPs and times (inference time) are calculated on 256×256 patch.

Variants	Fusion Strategy			w/o AFRM	PSNR	SSIM	Param(M)	FLOPs(G)	Time(ms)
	Sum	Cat	MSFM						
DDCAM $\rightarrow$ CA	$\checkmark$			$\times$	39.45	0.9805	11.91	17.61	142.30
		$\checkmark$		$\times$	39.76	0.9815	11.99	17.74	142.33
			$\checkmark$	$\times$	39.89	0.9833	12.03	17.74	142.88
	$\checkmark$			$\checkmark$	39.99	0.9822	12.04	18.07	170.36
		$\checkmark$		$\checkmark$	40.08	0.9827	12.13	18.22	170.77
			$\checkmark$	$\checkmark$	40.19	0.9830	12.16	18.22	171.03
DDCAM	$\checkmark$			$\times$	40.02	0.9820	12.25	17.95	145.69
		$\checkmark$		$\times$	40.14	0.9827	12.33	18.09	145.85
			$\checkmark$	$\times$	40.18	0.9829	12.36	18.09	146.68
	$\checkmark$			$\checkmark$	40.37	0.9835	12.38	18.42	174.07
		$\checkmark$		$\checkmark$	40.42	0.9836	12.46	18.55	174.33
			$\checkmark$	$\checkmark$	40.64	0.9848	12.50	18.55	174.77

Table 4

Ablation study of dual-branch architecture.

Model type	PSNR	SSIM	Param(M)	FLOPs(G)	Time(ms)
hybird	39.94	0.9815	11.83	17.58	138.09
dual-branch	40.64	0.9848	12.50	18.55	174.77

4.4.1. Effectiveness of dual-branch architecture

To validate the effectiveness of the dual-branch architecture in our DDNet, we conduct an ablation study by removing the decoder in the low-resolution branch while retaining the encoder exclusively. This configuration aims to make the low-resolution branch focus on extracting global contextual information from the input image. The encoded multi-scale features from the low-resolution encoder are then merged with the decoded high-resolution features from the high-resolution branch. Crucially, supervision is applied exclusively to the final high-resolution output to enforce end-to-end learning. A detailed illustration of this modified structure will be provided in the supplementary material, and we name it hybrid architecture. As shown in Table 4, removing the decoder from the low-resolution branch results in a significant performance drop, with PSNR and SSIM decreasing by 0.7 dB and 0.0033, respectively. The experimental results validate that our dual-branch architecture demonstrates superior capability in high-resolution image de-noising tasks.

4.4.2. Impact of the AFRM

To validate the effectiveness of the AFRM, we conduct experiments by incorporating or removing the AFRM under various ablation strategies. As presented in Table 3, the quantitative result achieved by our DDNet is shown in the last row, which exhibits the optimal performance. Conversely, the absence of the AFRM module (as demonstrated in the ninth row) leads to a notable performance degradation of 0.46 dB decreases in PSNR and 0.0019 decreases in SSIM, despite only a minimal decrease in the number of parameters. Furthermore, it can be clearly observed in Table 3 that comparative experiments conducted under other ablation strategies similarly validate the significant contribution of the AFRM to the model’s performance. To further validate the rationale behind the two-stage enhancement design of AFRM, we retain only the second-stage feature enhancement and remove the first-stage interaction, and the experimental results are reported in Table 5. Moreover, we replace our original adaptive mechanism with conventional frequency-decomposition schemes such as the Discrete Cosine Transform (DCT) and the Wavelet Transform, and the experimental results are summarized in Table 5. Evidently, our two-stage enhancement strat-

egy and adaptive frequency-decomposition approach achieves the best performance.

4.4.3. Impact of the MSFM

To demonstrate the effectiveness of our proposed MSFM, we conduct ablation studies by replacing MSFM with summation or channel concatenation, as indicated by “Sum” and “Cat” under “Fusion Strategy” in Table 3. The quantitative comparisons in Table 3 (grouped by three adjacent rows) demonstrate that our proposed Multi-Spectral Fusion Module (MSFM) achieves significantly higher metrics than both concatenation (Cat) and summation (Sum) fusion strategies. Taking the last two rows as an example, the PSNR and SSIM values decrease by 0.22 dB and 0.0012 with a minimal parameter reduction of 0.04M. This suggests that the absence of MSFM leads to an ineffective aggregation of frequency domain information, consequently degrading the evaluation metrics.

4.4.4. Effectiveness of the DDCAM

Similarly, Table 3 presents the experimental results obtained by replacing the DDCAM with a cross-attention (CA) module. By comparing the first six rows with the last six rows of the table, it is clearly observable that the utilization of DDCAM consistently yields significantly better performance than the model employing CA across various ablation strategies. Taking our method DDNet (last row) and the variant with only DDCAM replaced (sixth row) as an example, the PSNR and SSIM metrics decrease by 0.45 dB and 0.0018, respectively. This compellingly demonstrates that DDCAM can effectively align the distributions of the dual-branch features, thereby mitigating the negative impact caused by feature discrepancies. Moreover, the ablation study presented in Table 5 involves retaining only the cross-attention operation while removing the differential attention mechanism. The substantial degradation in metrics observed confirms that the proposed differential processing is essential for achieving optimal performance.

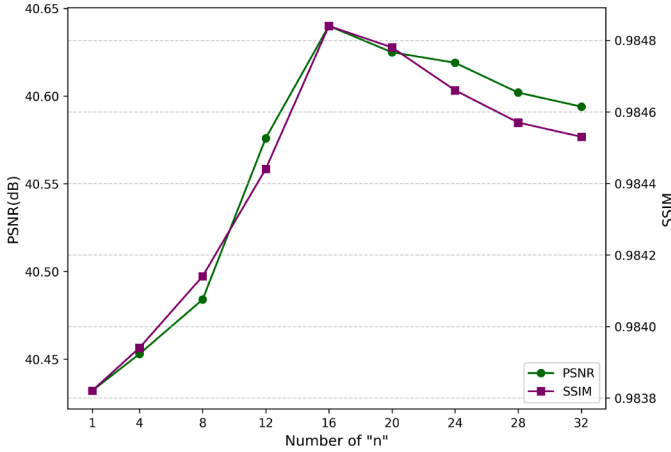
4.4.5. Selection of the value of n for MSFM

Table 6 and Fig. 9 illustrates the impact of varying the parameter n within the MSFM on the performance of model. Notably, when n is set to 1, our proposed frequency aggregation operation degenerates into Global Average Pooling (GAP). This results in the poorest performance, with PSNR and SSIM values 0.3dB and 0.0030 lower than our DDNet. When n is insufficiently small, a significant or complete loss of high-frequency information occurs, resulting in suboptimal model performance. As the value of n increases, the model’s performance gradually improves. However, when n is unduly large (e.g., 32), the MSFM incorporates an overabundance of high-frequency coordinates, causing the

Table 5

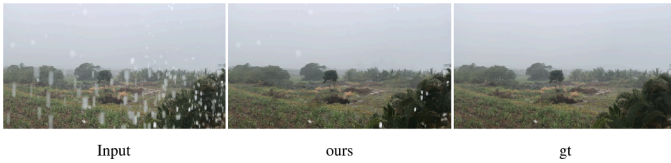
Ablation study on the key internal components within each module. AFRM (w/o 1st stage) represents the variant where only the second-stage feature enhancement is retained and the first-stage interaction is removed. AFRM (DCT) and AFRM (Wavelet) denote variants of AFRM in which the adaptive frequency extraction mechanism is replaced by DCT and wavelet transforms, respectively. "Remove 'Diff'" refers to the ablation of the differential attention module in DDCAM.

Methods	AFRM (w/o 1st stage)	AFRM (DCT)	AFRM (Wavelet)	Remove "Diff"	Ours
PSNR	40.17	40.28	40.21	40.45	40.64
SSIM	0.9826	0.9831	0.9835	0.9839	0.9848

**Fig. 9.** Performance curve of the number of n on RealRain1k-H (Li et al., 2022).**Table 6**

Ablation study on parameter n of MSFM.

Nmuber of n	1	4	8	16(ours)	32
PSNR	40.43	40.45	40.48	40.64	40.59
SSIM	0.9838	0.9839	0.9841	0.9848	0.9845

**Fig. 10.** A failure case where rain streaks exhibit extensive clustered and patchy patterns.

fused features to allocate disproportionate attention to high-frequency components whose semantic contribution is marginal. This redundancy in the frequency domain undermines feature consistency and ultimately produces a decline in model performance. It should be noted that this experiment was conducted on RealRain1k (Li et al., 2022), and the results on the LHP (Guo et al., 2023) and RainDirection (Liu et al., 2021) datasets are provided in the supplementary material, where experiments on these datasets demonstrate that the optimal value of n is consistently 16.

5. Limitations

Although our method achieves competitive performance in several scenarios, it also exhibits limitations under extreme conditions. As shown in Fig. 10, when processing images with dense rain streaks characterized by extensive clustered and patchy patterns, our method fails to completely remove all rain streaks. Additionally, certain background structures exhibit blurring or distortion. This limitation stems from the fact that the rain streaks in Fig. 10 exhibit extensive clustered and patchy patterns, which differ significantly from the general rain streak charac-

teristics, consequently leading to suboptimal performance in these specific and challenging scenarios.

6. Conclusion

In this paper, we propose a novel frequency-aware dual-branch deraining network (DDNet) for high resolution image deraining, which comprises a CNN and Transformer branch architecture along with three key modules. Specifically, the Adaptive Frequency Reciprocal Module (AFRM) is designed to interact and enhance frequency domain feature representations from the dual branches, while the Multi-Spectral Fusion Module (MSFM) effectively aggregates frequency domain information and dynamically fuses features. Finally, the Dual Differential Cross Attention Module (DDCAM) is proposed to mitigate the negative impact arising from discrepancies between the CNN and Transformer branches. Extensive experiments on multiple datasets demonstrate that DDNet achieves state-of-the-art performance while maintaining low parameter count and computational complexity.

Our future work will involve a deeper investigation into the extraction and fusion of spatial and frequency domain features, with the objective of developing more powerful and lightweight network architectures for high-resolution image deraining and broader high-resolution image restoration applications.

CRedit authorship contribution statement

Weilong Huang: Writing – original draft, Software, Methodology, Conceptualization; **Jiaxue Mei:** Validation, Data curation; **Tao Yan:** Writing – review & editing, Supervision; **Yinghui Wang:** Writing – review & editing, Supervision; **Xiaojun Chang:** Writing – review & editing, Investigation.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The work is supported by the [National Natural Science Foundation of China \(61902151\)](#).

Supplementary material

Supplementary material associated with this article can be found in the online version at [10.1016/j.neunet.2026.108936](https://doi.org/10.1016/j.neunet.2026.108936).

References

- Ahmed, N., Natarajan, T., & Rao, K. R. (1974). Discrete cosine transform. *IEEE Transactions on Computers*, C-23(1), 90–93.
- Ba, J. L., Kiros, J. R., & Hinton, G. E. (2016). Layer normalization. arXiv: 1607.06450.
- Benjdira, B., Ali, A. M., & Koubaa, A. (2023). Streamlined global and local features combinator (sglc) for high resolution image dehazing. In *Ieee conf. comput. vis. pattern recog. worksh.* (pp. 1855–1864).
- Bianco, S., Celona, L., Piccoli, F., & Schettini, R. (2019). High-resolution single image dehazing using encoder-decoder architecture. In *Ieee conf. comput. vis. pattern recog. worksh.* (pp. 1927–1935).
- Cai, H., Li, J., Hu, M., Gan, C., & Han, S. (2023a). Efficientvit: Lightweight multi-scale attention for high-resolution dense prediction. In *Int. conf. comput. vis.* (pp. 17256–17267).
- Cai, L., Fu, Y., Huo, W., Xiang, Y., Zhu, T., Zhang, Y., Zeng, H., & Zeng, D. (2023b). Multiscale attentive image de-raining networks via neural architecture search. *IEEE Transactions on Circuits and Systems for Video Technology*, 33(2), 618–633.
- Chen, C.-F. R., Fan, Q., & Panda, R. (2021). Crossvit: Cross-attention multi-scale vision transformer for image classification. In *Int. conf. comput. vis.* (pp. 347–356).
- Chen, H., Chen, X., Wu, C., Zheng, Z., Pan, J., & Fu, X. (2024a). Towards ultra-high-definition image deraining: A benchmark and an efficient method. arXiv: 2405.17074.
- Chen, L., Chu, X., Zhang, X., & Sun, J. (2022). Simple baselines for image restoration. In *Eur. conf. comput. vis.* (pp. 17–33).
- Chen, L., Fu, Y., Gu, L., Yan, C., Harada, T., & Huang, G. (2024b). Frequency-aware feature fusion for dense image prediction. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 46(12), 10763–10780.
- Chen, X., Li, H., Li, M., & Pan, J. (2023a). Learning a sparse transformer network for effective image deraining. In *Ieee conf. comput. vis. pattern recog.* (pp. 5896–5905).
- Chen, X., Pan, J., & Dong, J. (2024c). Bidirectional multi-scale implicit neural representations for image deraining. In *Ieee conf. comput. vis. pattern recog.* (pp. 25627–25636).
- Chen, X., Pan, J., Lu, J., Fan, Z., & Li, H. (2023b). Hybrid cnn-transformer feature fusion for single image deraining. In *Aaai conf. artif. intell.* (pp. 378–386).
- Cooley, J. W., Lewis, P. A. W., & Welch, P. D. (1969). The fast fourier transform and its applications. *IEEE Transactions on Education*, 12(1), 27–34.
- Dosovitskiy, A., Beyer, L., Kolesnikov, A., Weissenborn, D., Zhai, X., Unterthiner, T., Dehghani, M., Minderer, M., Heigold, G., Gelly, S., Uszkoreit, J., & Houlsby, N. (2021). An image is worth 16x16 words: Transformers for image recognition at scale. In *Int. conf. learn. represent.*
- Fu, X., Huang, J., Zeng, D., Huang, Y., Ding, X., & Paisley, J. (2017). Removing rain from single images via a deep detail network. In *Ieee conf. comput. vis. pattern recog.* (pp. 1715–1723).
- Gao, N., Jiang, X., Zhang, X., & Deng, Y. (2024a). Efficient frequency-domain image deraining with contrastive regularization. In *Eur. conf. comput. vis.* (pp. 240–257).
- Gao, T., Wen, Y., Zhang, K., Zhang, J., Chen, T., Liu, L., & Luo, W. (2024b). Frequency-oriented efficient transformer for all-in-one weather-degraded image restoration. *IEEE Transactions on Circuits and Systems for Video Technology*, 34(3), 1886–1899. <https://doi.org/10.1109/TCSVT.2023.3299324>
- Gu, S., Meng, D., Zuo, W., & Zhang, L. (2017). Joint convolutional analysis and synthesis sparse representation for single image layer separation. In *Int. conf. comput. vis.* (pp. 1717–1725).
- Guo, J., Han, K., Wu, H., Tang, Y., Chen, X., Wang, Y., & Xu, C. (2022). Cmt: Convolutional neural networks meet vision transformers. In *Ieee conf. comput. vis. pattern recog.* (pp. 12165–12175).
- Guo, Y., Xiao, X., Chang, Y., Deng, S., & Yan, L. (2023). From sky to the ground: A large-scale benchmark and simple baseline towards real rain removal. In *Int. conf. comput. vis.* (pp. 12097–12107).
- Hu, J., Shen, L., & Sun, G. (2018). Squeeze-and-excitation networks. In *Ieee conf. comput. vis. pattern recog.* (pp. 7132–7141).
- Huang, Z., Zhang, Z., Lan, C., Zha, Z.-J., Lu, Y., & Guo, B. (2023). Adaptive frequency filters as efficient global token mixers. In *Int. conf. comput. vis.* (pp. 6026–6036).
- Jiang, K., Jiang, J., Liu, X., Xu, X., & Ma, X. (2024). Fmnet: Image deraining via frequency mutual revision. In *Aaai conf. artif. intell.* (pp. 12892–12900).
- Jiang, K., Wang, Z., Yi, P., Chen, C., Huang, B., Luo, Y., Ma, J., & Jiang, J. (2020). Multi-scale progressive fusion network for single image deraining. In *Ieee conf. comput. vis. pattern recog.* (pp. 8343–8352).
- Kang, L.-W., Lin, C.-W., & Fu, Y.-H. (2012). Automatic single-image-based rain streaks removal via image decomposition. *IEEE Transactions on Image Processing*, 21(4), 1742–1755.
- Kawar, B., Elad, M., Ermon, S., & Song, J. (2022). Denoising diffusion restoration models. In *Adv. neural inf. process. syst.* (pp. 23593–23606).
- Laplanche, P. A., Cravey, R., Dunleavy, L. P., Antonakos, J. L., LeRoy, R., East, J., Buris, N. E., Conant, C. J., Fryda, L., Boyd, R. W. et al. (2018). Comprehensive dictionary of electrical engineering. CRC Press.
- LeCun, Y., Bottou, L., Bengio, Y., & Haffner, P. (1998). Gradient-based learning applied to document recognition. *Proceedings of the IEEE*, 86(11), 2278–2324.
- Li, S., Araujo, I. B., Ren, W., Wang, Z., Tokuda, E. K., Junior, R. H., Cesar-Junior, R., Zhang, J., Guo, X., & Cao, X. (2019). Single image deraining: A comprehensive benchmark analysis. In *Ieee conf. comput. vis. pattern recog.* (pp. 3833–3842).
- Li, W., Zhang, Q., Huang, J., Huang, Z., Tian, X., & Tao, D. (2022). Toward real-world single image deraining: A new benchmark and beyond. arXiv: 2206.05514.
- Li, X., Wu, J., Lin, Z., Liu, H., & Zha, H. (2018). Recurrent squeeze-and-excitation context aggregation net for single image deraining. In *Eur. conf. comput. vis.* (pp. 262–277).
- Li, Y., Tan, R. T., Guo, X., Lu, J., & Brown, M. S. (2016). Rain streak removal using layer priors. In *Ieee conf. comput. vis. pattern recog.* (pp. 2736–2744).
- Liu, Y., Li, D., Fu, X., Lu, X., Huang, J., & Zha, Z.-J. (2025). Uhd-processor: Unified uhd image restoration with progressive frequency learning and degradation-aware prompts. In *Ieee conf. comput. vis. pattern recog.* (pp. 23121–23130).
- Liu, Y., Yue, Z., Pan, J., & Su, Z. (2021). Unpaired learning for deep image deraining with rain direction regularizer. In *Int. conf. comput. vis.* (pp. 4733–4741).
- Loshchilov, I., & Hutter, F. (2017). SGDR: Stochastic gradient descent with warm restarts. In *Int. conf. learn. represent.*
- Lu, J., Batra, D., Parikh, D., & Lee, S. (2019). Vilbert: Pretraining task-agnostic visiolinguistic representations for vision-and-language tasks. *Advances in Neural Information Processing Systems*, 32, 13–23.
- Luo, W., Li, Y., Urtasun, R., & Zemel, R. (2016). Understanding the effective receptive field in deep convolutional neural networks. *Advances in Neural Information Processing Systems*, (pp. 4905–4913).
- Luo, Y., Xu, Y., & Ji, H. (2015). Removing rain from a single image via discriminative sparse coding. In *Int. conf. comput. vis.* (pp. 3397–3405).
- Luo, Z., Gustafsson, F. K., Zhao, Z., Sjölund, J., & Schön, T. B. (2023). Image restoration with mean-reverting stochastic differential equations. In *Int. conf. mach. learn.* (pp. 23045–23066).
- Magid, S. A., Zhang, Y., Wei, D., Jang, W.-D., Lin, Z., Fu, Y., & Pfister, H. (2021). Dynamic high-pass filtering and multi-spectral attention for image super-resolution. In *Int. conf. comput. vis.* (pp. 4268–4277).
- Mallat, S. G. (1989). A theory for multiresolution signal decomposition: The wavelet representation. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 11(7), 674–693.
- Nyquist, H. (1928). Certain topics in telegraph transmission theory. *Transactions of the American Institute of Electrical Engineers*, 47(2), 617–644.
- Özdenizci, O., & Legenstein, R. (2023). Restoring vision in adverse weather conditions with patch-based denoising diffusion models. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(8), 10346–10357.
- Park, N., & Kim, S. (2022). How do vision transformers work? In *Int. conf. learn. represent.*
- Qin, Z., Zhang, P., Wu, F., & Li, X. (2021). Fcanet: Frequency channel attention networks. In *Int. conf. comput. vis.* (pp. 763–772).
- Quan, R., Yu, X., Liang, Y., & Yang, Y. (2021). Removing raindrops and rain streaks in one go. In *Ieee conf. comput. vis. pattern recog.* (pp. 9143–9152).
- Ren, D., Zuo, W., Hu, Q., Zhu, P., & Meng, D. (2019). Progressive image deraining networks: A better and simpler baseline. In *Ieee conf. comput. vis. pattern recog.* (pp. 3932–3941).
- Shannon, C. E. (1949). Communication in the presence of noise. *Proceedings of the IRE*, 37(1), 10–21.
- Tay, Y., Dehghani, M., Bahri, D., & Metzler, D. (2022). Efficient transformers: A survey. *ACM Computing Surveys*, 55(6), 1–28.
- Touvron, H., Vedaldi, A., Douze, M., & Jégou, H. (2019). Fixing the train-test resolution discrepancy. *Advances in Neural Information Processing Systems*, (pp. 8252–8262).
- Vaswani, A., Shazeer, N., Parmar, N., Uszkoreit, J., Jones, L., Gomez, A. N., Kaiser, L., & Polosukhin, I. (2017). Attention is all you need. *Advances in Neural Information Processing Systems*, (pp. 6000–6010).
- Wallace, G. K. (1992). The JPEG still picture compression standard. *IEEE Transactions on Consumer Electronics*, 38(1), xviii–xxiv.
- Wang, C., Pan, J., Wang, W., Fu, G., Liang, S., Wang, M., Wu, X.-M., & Liu, J. (2024a). Correlation matching transformation transformers for uhd image restoration. In *Aaai conf. artif. intell.* (pp. 5336–5344).
- Wang, C., Yan, T., Huang, W., Chen, X., Xu, K., & Chang, X. (2025a). Apanet: Asymmetrical parallax attention network for efficient stereo image deraining. *IEEE Transactions on Computational Imaging*, 11, 101–115.
- Wang, Q., Jiang, K., Wang, Z., Ren, W., Zhang, J., & Lin, C.-W. (2024b). Multi-scale fusion and decomposition network for single image deraining. *IEEE Transactions on Image Processing*, 33, 191–204.
- Wang, Y., Yan, X., Niu, Y., Gong, L., Guo, Y., & Wei, M. (2025b). Rethinking mixture of rain removal via depth-guided adversarial learning. *Neural Networks*, 191, 107739.
- Wen, Y., Gao, T., Chen, T., Li, Z., Liu, M., & Liu, L. (2025a). Cross-level interaction and intralevel fusion network for remote sensing image dehazing. *IEEE Transactions on Geoscience and Remote Sensing*, 63, 1–14.
- Wen, Y., Gao, T., Li, Z., Zhang, J., Zhang, K., & Chen, T. (2025b). All-in-one weather-degraded image restoration via adaptive degradation-aware self-prompting model. *IEEE Transactions on Multimedia*, 27, 3343–3355.
- Wen, Y., Gao, T., Zhang, J., Li, Z., & Chen, T. (2023). Encoder-free multiaxis physics-aware fusion network for remote sensing image dehazing. *IEEE Transactions on Geoscience and Remote Sensing*, 61, 1–15.
- Xiao, J., Fu, X., Liu, A., Wu, F., & Zha, Z.-J. (2022). Image de-raining transformer. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 45(11), 12978–12995.
- Xiao, Y., & Xia, Y. (2025). Pixel adaptive deep-unfolding neural network with state space model for image deraining. *Neural Networks*, 191, 107845.
- Xu, S., Ke, Q., Peng, J., Cao, X., & Zhao, Z. (2024). Pan-denoising: Guided hyperspectral image denoising via weighted represent coefficient total variation. *IEEE Transactions on Geoscience and Remote Sensing*, 62, 1–14.
- Xu, S., Zhao, Z., Cao, X., Peng, J., Zhao, X.-L., Meng, D., Zhang, Y., Timofte, R., & Van Gool, L. (2025). Parameterized low-rank regularizer for high-dimensional visual data. *International Journal of Computer Vision*, (pp. 1–24).
- Yan, T., Zhu, X., Chen, X., He, W., Wang, C., Yang, Y., Wang, Y., & Chang, X. (2024). Glgfn: Global-local grafting fusion network for high-resolution image deraining. *IEEE Transactions on Circuits and Systems for Video Technology*, 34(11), 10860–10873.
- Yang, W., Tan, R. T., Feng, J., Liu, J., Guo, Z., & Yan, S. (2017). Deep joint rain detection and removal from a single image. In *Ieee conf. comput. vis. pattern recog.* (pp. 1685–1694).

- Yasarla, R., & Patel, V. M. (2019). Uncertainty guided multi-scale residual learning-using a cycle spinning cnn for single image de-raining. In *Ieee conf. comput. vis. pattern recog.* (pp. 8397–8406).
- Ye, T., Dong, L., Xia, Y., Sun, Y., Zhu, Y., Huang, G., & Wei, F. (2025). Differential transformer. In *Int. conf. learn. represent.*
- Zamir, S. W., Arora, A., Khan, S., Hayat, M., Khan, F. S., & Yang, M.-H. (2022). Restormer: Efficient transformer for high-resolution image restoration. In *Ieee conf. comput. vis. pattern recog.* (pp. 5718–5729).
- Zamir, S. W., Arora, A., Khan, S., Hayat, M., Khan, F. S., Yang, M.-H., & Shao, L. (2021). Multi-stage progressive image restoration. In *Ieee conf. comput. vis. pattern recog.* (pp. 14816–14826).
- Zhang, H., & Patel, V. M. (2018). Density-aware single image de-raining using a multi-stream dense network. In *Ieee conf. comput. vis. pattern recog.* (pp. 695–704).
- Zhang, R., Yu, J., Chen, J., Li, G., Lin, L., & Wang, D. (2024). A prior guided wavelet-spatial dual attention transformer framework for heavy rain image restoration. *IEEE Transactions on Multimedia*, 26, 7043–7057.
- Zhao, C., Cai, W., Hu, C., & Yuan, Z. (2024). Cycle contrastive adversarial learning with structural consistency for unsupervised high-quality image deraining transformer. *Neural Networks*, 178, 106428.
- Zhao, C., Chen, Z., Xu, Y., Gu, E., Li, J., Yi, Z., Wang, Q., Yang, J., & Tai, Y. (2025). From zero to detail: Deconstructing ultra-high-definition image restoration from progressive spectral perspective. In *Ieee conf. comput. vis. pattern recog.* (pp. 17935–17946).
- Zheng, Z., Ren, W., Cao, X., Hu, X., Wang, T., Song, F., & Jia, X. (2021). Ultra-high-definition image dehazing via multi-guided bilateral learning. In *Ieee conf. comput. vis. pattern recog.* (pp. 16180–16189). IEEE.
- Zhou, W., Sun, F., Jiang, Q., Cong, R., & Hwang, J.-N. (2023a). Wavenet: Wavelet network with knowledge distillation for RGB-t salient object detection. *IEEE Transactions on Image Processing*, 32, 3027–3039.
- Zhou, W., Zhang, H., Yan, W., & Lin, W. (2023b). Mmsmcnet: Modal memory sharing and morphological complementary networks for rgb-t urban scene semantic segmentation. *IEEE Transactions on Circuits and Systems for Video Technology*, 33(12), 7096–7108.
- Zhou, W., Zhu, Y., Lei, J., Wan, J., & Yu, L. (2022). Ccafnet: Crossflow and cross-scale adaptive fusion network for detecting salient objects in rgb-d images. *IEEE Transactions on Multimedia*, 24, 2192–2204.
- Zhu, L., Fu, C.-W., Lischinski, D., & Heng, P.-A. (2017). Joint bi-layer optimization for single-image rain streak removal. In *Int. conf. comput. vis.* (pp. 2545–2553).
- Zou, X., Xiao, F., Yu, Z., Li, Y., & Lee, Y. J. (2023). Delving deeper into anti-aliasing in convnets. *International Journal of Computer Vision*, 131(1), 67–81.